Repository notes for *Chromatically Constrained Tensor Graphs: Concurrency and Connectome Topology*

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These notes contain the detailed construction procedures, parameter accounting, null-model definitions, extended graph audits, and computational-model specifications underlying the main article. The main text now emphasizes two comparatively simple constructions: the local prune–repair construction and the 8,000-node seed-encoded construction. These notes give reproducibility details for those models and retain the earlier, more elaborate 16,807-node fixed-partition relocation pipeline as an optional refinement for the case in which fiber-bundle multiplicity is itself treated as an explicit target. The organization separates reproducibility-oriented methods from extended results that are summarized more compactly in the main text.

S1 Supplementary Methods

S1.1 Local construction and quotienting

In this section we show that a simple two-part algorithm with a small number of parameters can produce a graph that matches numerous properties of the connectome reported across the literature.

The main article next introduces a second, simpler construction in which a four-block mesoscale template is encoded directly in a 20-vertex seed and propagated by tensor growth. We give its exact reproducibility details below. A later supplementary subsection then retains the older, more elaborate fixed-partition relocation pipeline. That pipeline is no longer the principal second construction; it is included because it demonstrates how the fiber-bundle distribution can be matched when that statistic is targeted explicitly.

A small tripartite seed is expanded repeatedly by tensor product. After each expansion, the graph is pruned, locally repaired, and closed by chromatically legal triangle additions. Finally, the resulting parent graph is coarsened to a 1,015-node quotient.

Algorithm 1A: Parent construction by local repair

Input. A tripartite seed graph $S = (V_S, E_S)$ with proper 3-coloring $c_S : V_S \to \{0, 1, 2\}$; tensor depth K; minimum degree $d_{\min}$; pruning fraction p; triangle-addition fraction a; and a fixed pseudorandom order, which sets the order in which repair attachments are tried and wedge centers are scanned in the closure step. Different orders give different members of a family of

parents with the same guarantees (see the main text for their effect).

Definitions. At scale k, let $n_k = |V(G_k)|$ and define the geometric-mean edge budget

$$b_k = \text{round}\left(n_k \sqrt{\frac{n_k - 1}{2}} \right).$$

A legal edge is an edge $\{x, y\}$ satisfying $c(x) \neq c(y)$. Only legal edges are added; deletions are unrestricted. The triangle support of an edge $\{u, v\}$ is $|N(u) \cap N(v)|$.

Initialize. $G_0 = S$, $c = c_S$.

For $k = 0, 1, \ldots, K - 1$, perform the following steps.

1. **Tensor expansion.** Set $\widetilde{G}_{k+1} = G_k \otimes S$. The vertices of $\widetilde{G}_{k+1}$ are pairs (u, a), where $u \in V(G_k)$ and $a \in V_S$. An edge joins (u, a) and (v, b) exactly when $\{u, v\} \in E(G_k)$ and $\{a, b\} \in E_S$. Project the coloring according to $c((u, a)) = c(u)$. Thus every tensor edge remains cross-color, and the proper 3-coloring is preserved. Because G_k and S are connected and non-bipartite, $\widetilde{G}_{k+1}$ is also connected and non-bipartite; record one odd cycle C_{k+1} before pruning as an invariant witness.

2. **Community-aware pruning.** Score every edge of $\widetilde{G}_{k+1}$ by its triangle support. Retain the round$(p\, b_{k+1})$ highest-support edges and delete the remaining edges.

3. **Local chromatic repair.** For each vertex whose degree is below $d_{\min}$, add legal cross-color edges through the vertex's two-hop neighborhood whenever possible, preferentially selecting edges that close local triangles. If the local candidate pool is exhausted, use any available legal endpoint. If the graph is disconnected, add legal bridging edges until connectivity is restored.

4. **Chromatic triangle closure.** Admit the closing edge $\{u, w\}$ of an open wedge $u - v - w$ as a candidate only when $c(u) \neq c(w)$. Scan wedge centers in the fixed pseudorandom order (a center of degree above 60 contributes a random sample of 60 of its neighbors), score each candidate pair by the number of scanned centers adjacent to both of its endpoints, and stop once five times the needed number of candidates has been collected. Add the highest-scoring candidates until the total edge count reaches round$(a\, b_{k+1})$. At the final scale the admitted closures average 15.5 common neighbors, against 19.4 for an exact ranking of all legal open wedges.

5. **Invariant safeguard.** Verify connectivity and non-bipartiteness of the edited graph. If it is disconnected, add legal cross-color bridging edges until connectivity is restored. If it is bipartite, restore any tensor edges of the recorded odd cycle C_{k+1} that were deleted during this stage until the whole cycle is present again. These restored tensor edges are cross-color under the projected coloring. The invariant safeguard takes precedence over the nominal edge-count target.

Set G_{k+1} equal to the resulting graph. After K tensor steps, G_K is the parent graph.

Output. The locally repaired parent graph G_K, together with its proper 3-coloring c.

Algorithm 1B: Coarsening to the quotient

Input. The parent graph G_K produced by Algorithm 1A; Louvain resolution γ; target quotient size N; and a fixed random seed.

Definitions. For two communities A and B, let $w(A, B)$ denote the number of edges of G_K with one endpoint in A and the other in B. Define their intercommunity edge density by $\rho(A, B) = w(A, B)/(|A|\,|B|)$.

1. **High-resolution partition.** Run Louvain community detection on G_K at resolution γ, chosen so that the number of detected communities exceeds the target N.
2. **Density-based merging.** While the number of communities is greater than N, choose a smallest community A. Among the communities adjacent to A, select a community B maximizing $\rho(A, B)$, and merge A into B. Continue until exactly N supernodes remain.
3. **Quotient construction.** Construct the quotient graph H_{LR} with one vertex for each surviving supernode. Two distinct supernodes are adjacent in H_{LR} if and only if at least one edge of G_K has endpoints in the two corresponding communities.

Output. The local quotient H_{LR}.

For the concrete run reported here, the seed has $|V_S| = 7$ vertices and $|E_S| = 9$ edges. We use tensor depth $K = 4$, minimum degree $d_{\min} = 3$, pruning fraction $p = 0.10$, triangle-addition fraction $a = 0.18$, Louvain resolution $\gamma = 40$, target quotient size $N = 1{,}015$, a fixed pseudorandom order for repair and closure, and Louvain random seed 42. With these values, the local construction produces a quotient H_{LR} with

$$n = 1{,}015, \qquad |E| = 67{,}094, \qquad \rho = 0.1304, \qquad Q = 0.524.$$

This is the whole local construction. It has no empirical adjacency matrix, no anatomical coordinates, no tractography weights, no fitted edge probabilities, no later fixed-partition inverse-relocation stage. It is a local growth-and-repair mechanism followed by quotienting. Only node count, edge count, modularity, and low-degree nodes were used as targets.

Edge-budget normalization. The local construction was initially parameterized from the side of graph complexity. For a graph on n vertices, the sparsest connected graph has $O(n)$ edges, while the complete graph has $n(n-1)/2$ edges. Many graph complexity measures are small near both extremes: trees and paths are too regular, while complete graphs are also highly regular. The natural intermediate scale is therefore the geometric mean between the sparse connected scale and the complete-graph scale,

$$E_{\mathrm{gm}} = n\sqrt{\frac{n-1}{2}},$$

up to the immaterial distinction between using n and $n-1$ for the sparse connected reference. This is the edge scale associated with maximal graph complexity in several graph-complexity measures [1].

In the local construction we did not use the full complexity scale. We used a fraction of it, $E_{\mathrm{LR}} = \alpha E_{\mathrm{gm}}$ with $\alpha = 0.18$. At the final parent size $n = 16{,}807$, this gives $E_{\mathrm{LR}} = 277{,}319$. Thus the local graph lies well below the maximal-complexity scale. This was empirically necessary: pushing the budget closer to E_{gm} produced graphs that were too dense, too modular, and lost the low-degree periphery seen in the Budapest graph.

The global construction uses the same sparse regime but expresses it from the opposite direction. Instead of taking a fraction down from the maximal-complexity scale, it starts from the connectivity scale, of order $n \log n$, and works upward by a controlled multiple. At $n = 16{,}807$, the local budget 277,319 is already only $1.18\, n \log_2 n$, or about 3.39 times the classical connectivity threshold $n \ln n/2$.

S1.2 Eight-thousand-node seed-encoded construction

The second main construction is generated by a standalone procedure that does not read an empirical adjacency matrix or any other graph file. It begins from a hand-specified seed with 20 vertices and 31 edges. Each seed vertex has an independent color label and block label. The color classes have sizes 6/6/8; the blocks A, B, C, D have sizes 6/6/4/4. Blocks A, C form one larger group and B, D the other. Every seed edge is cross-color. The edge inventory is 26 within-block edges, four same-group cross-block edges, and one edge between the two larger groups.

Growth uses tensor product with in-loop triangle closure:

$$G_1 = S + A_0, \qquad G_2 = (G_1 \otimes S) + A_1, \qquad G_3 = (G_2 \otimes S) + A_2,$$

where A_0, A_1, A_2 are the legal closures made at the seed stage and at scales two and three. The final parent therefore has $20^3 = 8{,}000$ vertices. Closure is applied with quotas

$$\alpha = (3, 4, 30)$$

and degree-weighting exponent

$$\beta = 1.5.$$

For one candidate closure, a center vertex v is sampled with probability proportional to $\deg(v)^\beta$ and two neighbors u, w are selected. The edge $\{u, w\}$ is added if and only if

$$c(u) \neq c(w), \qquad b(u) = b(w), \qquad \{u, w\} \notin E.$$

No edge is removed at any stage. Selection inside the legal candidate pool is unranked. This is deliberate: ranking all closures by common-neighbor support was tested and concentrated additions too strongly in the densest local neighborhoods, reducing blind block recovery.

The finished parent contains

$$n = 8{,}000, \qquad m = 477{,}584,$$

with color classes 2,400/2,400/3,200 and blocks 2,400/2,400/1,600/1,600. Since tensor edges inherit the seed's cross-color legality and every closure is filtered explicitly, no chromatic repair step is required. The graph has zero monochromatic edges and contains a triangle, so

$\chi = 3$ exactly. Topology-only Louvain recovery of the finished parent at resolution $\gamma = 0.2$ (construct.py, Louvain seed 7; seeds 1, 2, 3 and 42 give the same partition) returns the four blocks with NMI = ARI = 1.000. Higher resolutions subdivide the blocks without merging across them: at $\gamma = 0.5$ the result depends on the seed (four to six communities), and the standard $\gamma = 1$ gives eight communities, each contained in a single block.

Coarsening is performed independently within each block. Louvain is run at high resolution; the adjacent pair of communities with the largest intercommunity density is then repeatedly merged until the target counts

$$307/306/201/201$$

are reached. These four target counts are the only numerical quantities taken from the Budapest reference in this construction. Supernodes are joined in the quotient whenever at least one parent edge crosses between them; no quotient thinning follows. The strict quotient therefore contains

$$n = 1{,}015, \qquad m = 64{,}760.$$

The simple construction deliberately leaves one important mismatch. If W_{AB} denotes the number of fine edges underlying quotient relation A–B, then the mean bundle multiplicity is 6.885 rather than Budapest's 2.033, and the Gini coefficient is 0.661 rather than 0.305. A sweep over the final closure quota shows that thinning the parent cannot remove this tail without simultaneously damaging block recovery. This is the motivation for the optional complex refinement below.

S1.3 Optional complex fiber-bundle refinement: integrated parent, fixed causal partition, and inverse relocation

Exploratory; not part of the main article's reported results.

This section gives the implementation-level construction used for the integrated parent and the final inverse quotient. The crucial distinction from the earlier forward-only description is that the reported inverse run does *not* optimize a new coarse map. The original integrated causal-dependency graph is clustered once, reduced to exactly 1,015 supernodes, and that assignment is then held fixed while fine causal edges are reorganized beneath it.

A key tool used throughout the construction is Louvain community detection, a modularity-optimization heuristic for partitioning a graph into communities [2]. Here it is used algorithmically. Because it is applied directly to the fine causal-dependency graph, the resulting supernodes are mesoscale populations derived from the organization of that causal substrate; no anatomical identity is assigned to them.

Seed graph. The integrated construction begins from the same fixed tripartite seed graph used in the local construction,

$$S = (V_S, E_S), \qquad |V_S| = 7, \qquad |E_S| = 9,$$

with colors

$$(0, 0, 1, 1, 2, 2, 2)$$

and seed edges

$$(0,2),(0,5),(1,3),(1,4),(1,6),(2,5),(2,6),(3,4),(3,6).$$

Every seed edge joins vertices of different colors.

Stage A: chromatic tensor-product expansion and pruning. Four tensor-product expansion steps are applied after the initial seed, giving

$$n = 7^{4+1} = 16{,}807$$

vertices. A product vertex (u, a) inherits the color of its parent coordinate,

$$c_k(u, a) = c_{k-1}(u),$$

so every tensor edge remains cross-color. Pruning is performed after each expansion. At current size n, the reported edge budget is

$$B(n) = \mathrm{round}\big(n \log_2(n) m_{\mathrm{mult}}\big), \qquad m_{\mathrm{mult}} = 1.0.$$

A Louvain partition is computed with seed 42 and resolution 1.0. Each edge receives score

$$s_A(u, v) = T(u, v) + 10^{-7}(\deg u + \deg v), \qquad T(u, v) = |\Gamma(u) \cap \Gamma(v)|,$$

and the budget is split between inter-community and intra-community edges with inter-community fraction $\eta = 0.35$. Highest-scoring edges are retained within each pool. Minimum-degree repair then inserts chromatically legal edges until every vertex has degree at least three, and disconnected components, if any, are reconnected by legal cross-color edges.

After the four expansion steps and repairs, an additional

$$1.2n = 20{,}168$$

chromatically legal triangle-closing edges are added. The resulting graph has 16,807 vertices and 273,695 edges.

Stage B: multiscale-local cross-community bridges. The box-covering step is used as a multiscale locality filter, inspired by network box-covering methods [3, 4]. For

$$l_B \in \{3, 4, 5, 6, 7\},$$

randomized greedy coverings are constructed using $\mathrm{dist}(c, x) < l_B$. Multiple vertex orderings are tried and the covering with the fewest boxes is retained. If $b_{l_B}(u)$ is the box label of vertex u, define

$$\kappa(u) = \big(b_3(u), b_4(u), b_5(u), b_6(u), b_7(u)\big).$$

A candidate bridge (u, v) is allowed only when

$$\kappa(u) = \kappa(v), \qquad \mathrm{comm}(u) \neq \mathrm{comm}(v), \qquad c(u) \neq c(v), \qquad (u, v) \notin E.$$

Valid candidates are shuffled with seed 42, and up to $84{,}000 \approx 5n$ are inserted. The reported run reaches 357,695 edges at this stage.

Stage C: multiscale-targeted sparsification. The bridge-enriched graph is sparsified while monitoring a descriptive box-covering slope. The internal stopping target is

$$d_B \leq 3.8.$$

For this stage the membership rule is $\mathrm{dist}(c, x) \leq l_B$ and the tested scales are

$$l_B \in \{2, 3, 4, 5, 6, 8, 10, 12\},$$

restricted to scales no larger than the measured diameter. Edges are ranked for removal by

$$s_C(u, v) = 100T(u, v) + \frac{1}{\deg(u) + \deg(v) + 1}.$$

Graph bridges are protected at the start of each sparsification round, and a removal is skipped if it would reduce either endpoint below degree two. Up to 20,000 removals are attempted per round, with edge-count floor $m_{\min} = 100{,}000$. The resulting original integrated parent has

$$n = 16{,}807, \qquad m = 337{,}983,$$

and remains properly three-colored.

Stage D: fixed community-derived causal partition. Louvain community detection is applied directly to the original integrated parent at resolution

$$\gamma_{\text{init}} = 20.0$$

with seed 42, initially producing approximately 1,054 communities. These are reduced to exactly

$$N = 1{,}015$$

by deterministic density-based merging. At each step, a smallest community C_i is merged into the adjacent community C_j maximizing

$$\frac{\mathrm{edges}(C_i, C_j)}{|C_i|\,|C_j|}.$$

If C_i has no external neighbor, it is merged into the smallest other community. Let

$$\pi : V_{\text{parent}} \to \{1, \ldots, 1015\}$$

denote the resulting assignment, and define the member set of supernode a by

$$V_a = \{u : \pi(u) = a\}.$$

This map is fixed throughout all reported inverse stages. Every fine vertex therefore remains in the same causally derived supernode while the fine edge set changes.

For any fine graph under this fixed map, define the complete cross-supernode bundle

$$E_{ab} = \{\{u, v\} \in E : \{\pi(u), \pi(v)\} = \{a, b\}\}, \qquad a \neq b,$$

with multiplicity $W_{ab} = |E_{ab}|$. The reported quotient is always reconstructed by complete counting of these sets.

Forward strict-coarsening obstruction. Strict collapse of the unchanged integrated parent under this map yields 143,679 cross-supernode relations, with modularity $Q = 0.124$, clustering $C = 0.373$, and characteristic path length 1.725. A sweep of 25 additional strict coarsenings, all forced to 1,015 groups, includes Leiden RB and CPM, balanced METIS, Louvain–METIS hybrids, nested maps, box-covering maps, and a random balanced control. The best modularity observed is 0.288, still below Budapest's 0.5574. The fixed-parent forward problem therefore does not reach the target mesoscale regime.

Mesoscale target-support scaffold. The first inverse stage uses a stored target support set $\mathcal{S}$ built from the weighted witness graph of the original integrated parent. This is a construction scaffold, not a valid strict quotient of the unchanged parent. Unweighted Louvain at resolution 0.95 gives four large blocks of sizes

$$389, \quad 265, \quad 239, \quad 122.$$

All witness relations within each top-level block are retained. Finer within-block partitions are computed with

block	γ_{L1}	γ_{L2}
0	0.9	1.0
1	0.7	1.0
2	1.0	2.0
3	1.0	2.0

and cross-top-block $L1$-group pairs are ranked by total witness weight. The ten strongest cross-block patches are used, with individual relations added in decreasing witness-weight order toward

$$\left\lfloor \frac{139N}{2} \right\rfloor.$$

The stored scaffold contains 70,538 relations. It is used only as the target support for the fine-edge redistribution below.

Stage E: bundle-weight redistribution. Starting from the original integrated parent, every current cross-supernode pair absent from $\mathcal{S}$ loses all of its fine support. Let the pairs in $\mathcal{S}$ be sorted in ascending order of their original witness multiplicity. The implementation samples the same number of Budapest `fiber_count_mean` values without replacement using seed 42, sorts the sampled values, and assigns them rank-for-rank. The integer target multiplicity is

$$t_p = \max\{1, \mathrm{round}(b_p)\}.$$

For an overweight pair, supporting fine edges are scored by common-neighbor support and the lowest-support edges are removed first. For an underweight pair (a, b), absent cross-color edges with $u \in V_a$ and $v \in V_b$ are ranked by common-neighbor support and the highest-scoring candidates are added until the target multiplicity is reached.

Fine vertices below degree three are repaired first through legal two-hop candidates ranked by common-neighbor support and then, if needed, by seeded random legal non-neighbors. Connectivity is restored by legal cross-color edges if required. Any remaining

parent-edge deficit is then restored inside supernodes, processing the largest supernodes first and preferring absent cross-color pairs with high common-neighbor support. The complete strict quotient recomputed from this parent has 71,696 relations and already passes the nine-criterion quotient audit.

Stage F: mesoscale support refinement. The complete strict quotient of the bundle-redistributed parent is partitioned by unweighted Louvain at resolution 0.95. Within the four largest blocks, additional unweighted partitions use resolutions $0.9, 0.7, 1.0, 1.0$. Existing relations lying in the same top-level block but different finer groups are ordered by ascending bundle multiplicity, and the first 6,500 are selected. Removing a selected coarse relation means removing *all* fine parent edges supporting it.

Replacement candidates are quotient nonrelations whose endpoints lie in the same finer group. They are ranked by coarse common-neighbor count and the top 6,500 are selected. For each selected pair (a, b), the implementation searches only the first 30 stored members of V_a and the first 30 stored members of V_b. Among legal missing cross-color edges in this 30×30 set, the candidate maximizing parent common-neighbor support is inserted when one exists. The fine parent is authoritative: only relations with explicit fine support survive into the final stage.

Stage G: final degree-profile adjustment. The final adjustment starts from the parent produced above and uses NumPy seed 11. The complete strict quotient is reconstructed under the same fixed map. The Budapest degree sequence is sorted, and generated supernodes are sorted by their current degree; target degrees are then assigned rank-for-rank. No anatomical node correspondence is implied.

Supernodes above target are processed from largest surplus to smallest. Incident quotient relations are considered in ascending bundle multiplicity, and a relation is removed only when its partner is also above target. When a relation is removed, all of its supporting fine edges are deleted. These removed fine edges define the budget available for new cross-supernode support.

Supernodes below target are processed from largest deficit to smallest. Each search attempt samples 60 supernode indices with replacement. The first sampled partner that is distinct, not already adjacent in the quotient, and also below target is selected. Stored member lists are scanned in order for chromatically legal missing fine edges; the implementation can insert up to two fine edges supporting the new coarse relation. Any remaining parent-size deficit is restored inside supernodes. Bundle-rank assignment is not rerun at this stage; final bundle statistics are measured from the completed parent.

Approximately 2,146 fine-edge changes occur in this final degree-profile adjustment. The completed relocation substrate has

16,807 vertices, 337,980 edges,

and its single final strict quotient has

1,015 vertices, 71,098 edges.

Parameter	Role
$K = 4$	Tensor-product expansion steps.
$m_{\text{mult}} = 1.0$	Multiplier in $B(n) = n \log_2(n) m_{\text{mult}}$.
$\eta = 0.35$	Inter-community fraction of the pruning budget.
$\gamma_{\text{prune}} = 1.0$	Louvain resolution during tensor-step pruning.
$d_{\min}^{\text{repair}} = 3$	Fine-vertex degree floor after pruning and during bundle repair.
$\alpha_{\triangle} = 1.2$	Post-expansion triangle-closing budget.
$\mathcal{L}_{\text{bridge}} = \{3, 4, 5, 6, 7\}$	Box scales used for bridge candidates.
$\alpha_{\text{bridge}} = 5$	Maximum bridge budget in units of n.
$d_B^* = 3.8$	Descriptive box-covering target during sparsification.
$\mathcal{L}_{\text{sparse}} = \{2, 3, 4, 5, 6, 8, 10, 12\}$	Scales used during sparsification.
$N = 1{,}015$	Number of fixed supernodes.
$\gamma_{\text{init}} = 20.0$, seed 42	Original-parent community partition.
$\gamma_0 = 0.95$	Coarsest hierarchy resolution.
10	Number of strongest cross-block $L1$-pair patches retained in the target scaffold.
seed 42	Bundle-value sampling and bundle-stage repairs.
without replacement	Sampling rule for Budapest bundle values.
400	Largest-component vertices scanned during connectivity repair.
6,500	Relations selected in mesoscale support refinement.
30	Stored members examined per supernode when realizing a refinement-stage relation.
seed 11	Final degree-profile adjustment.
60	Candidate partner supernodes sampled per deficit-search attempt.
$\approx 2{,}146$	Fine-edge changes in the final degree-profile adjustment.

Table S1: Parameters and deterministic search limits used in the reported integrated and inverse construction.

Reported-run parameters. The node count, coarse edge scale, degree distribution, and bundle-weight distribution are explicit constraints on the inverse solution. No individual Budapest node, edge, or tract is fitted anatomically.

S1.4 Eight-thousand-node recurrent image-inpainting architecture

The main article's current computational realization uses the 8,000-node parent rather than the complex relocation substrate. The task is 256×256 CelebA-HQ image inpainting. A convolutional encoder takes RGB plus a binary mask and produces six feature maps:

$$64 \times 256 \times 256, \quad 128 \times 128 \times 128, \quad 256 \times 64 \times 64,$$

$$512 \times 32 \times 32, \quad 512 \times 16 \times 16, \quad 512 \times 8 \times 8.$$

All six encoder levels are sent to the graph. A collector pools and projects them to channel width $C = 32$ and assigns exactly 2,400 feature slots to the first color class. On the 8,000-node substrate the slot counts are $512, 128, 32, 1{,}024, 480, 224$ for encoder levels $s_3, s_4, s_5, s_2, s_1, s_0$, respectively, summing exactly to 2,400.

Every fine graph edge supplies a fixed sparse routing location whose scalar weight is learned. Five sparse operators are used:

$$1 \to 2, \qquad 1 \to 3, \qquad 2 \to 3, \qquad 3 \to 2, \qquad 3 \to 1.$$

The two feedback masks are transposes of the corresponding forward masks but carry independent trainable weights. The model executes five recurrent passes. Population a_2 is computed from the old a_1 plus delayed $a_3 \to a_2$ feedback; a_1 is then re-anchored to the encoder-derived base state plus $a_3 \to a_1$ feedback; a_3 is recomputed from the fresh a_1 and current a_2. This produces a recurrent rather than feed-forward use of the three color classes.

Same-phase coactivation is modeled separately by shared linear-attention blocks operating within each non-empty (color, supernode) group. Attention therefore mixes nodes inside a causal phase without introducing same-color edges into the structural graph. The decoder readout uses all 3,200 nodes of the third color class. Learned projections over all third-class nodes produce the two deepest decoder inputs (8×8 and 16×16), while four exact node blocks from the final recurrent pass produce the remaining multiscale graph features. The graph replaces the ordinary U-Net bottleneck and the two deepest skips; the three shallow convolutional skips remain and are modulated by graph-derived features.

The graph-plus-attention model contains 74,911,834 parameters on the 8,000-node substrate, compared with 87,103,619 for the matched U-Net baseline. The graph's five sparse operators account for 806,871 trainable parameters and the same-phase attention blocks for 12,864.

The final comparison uses three independently trained seeds per model under the same train/validation split, mask generator, optimizer, and checkpoint rule; the evaluation protocol and results are reported in the main article.

S1.5 Auxiliary image-inpainting results on the 16,807-node relocation substrate

Exploratory; not part of the main article's reported results.

This subsection is retained from the earlier complex-substrate experiment. It is useful as an auxiliary demonstration that the larger relocation graph can be trained end to end, but it is no longer the primary computational experiment in the main article. The primary architecture now uses the 8,000-node substrate described immediately above.

Task and data. We evaluate the graph module in a 256×256 CelebA-HQ image-inpainting task. Each example contains an image x, a binary mask m, and the masked observation

$$x_{\text{obs}} = m \odot x.$$

The model receives the four-channel tensor $[x_{\text{obs}}, m]$ and predicts a raw reconstruction $\hat{x}$. The reported composite preserves the observed pixels,

$$\hat{x}_{\text{final}} = m \odot x_{\text{obs}} + (1 - m) \odot \hat{x}.$$

Training uses 28,000 images with 2,000 held out. The same irregular-mask routine is used for training and evaluation, but with different target-coverage distributions. During training, the target coverage is sampled from

$$U(0.09, 0.49).$$

Across 30,000 generated training masks, the achieved missing fractions span approximately

$$[0.0904, 0.6040].$$

The broader benchmark instead samples target coverage from

$$U(0.02, 0.70).$$

Of the 4,893 benchmark masks entering the scored coverage buckets, 579 (11.8%) fall outside the achieved training range. All of these lie below the training floor and within the 0–20% coverage band; the 20–40% and 40–60% bands lie within the achieved training range.

The benchmark-mask distribution is the primary reported evaluation because it is the protocol used for comparison with the matched U-Net baseline. Evaluation on the training-mask distribution is reported as a robustness check. Aggregates from the two mask distributions are not interpreted as a direct performance comparison because the distributions populate the six coverage buckets differently.

Architecture and graph substrate. The recurrent model operates directly on the final substrate, obtained after bundle-matching relocation, mesoscale support refinement, and degree matching:

$$n = 16{,}807, \qquad m = 337{,}980, \qquad \chi = 3,$$

with fixed color-class sizes

$$n_1 = 4{,}802, \qquad n_2 = 4{,}802, \qquad n_3 = 7{,}203.$$

The fixed 1,015-supernode assignment is retained as the mesoscale grouping used by the optional supernode-attention modules. There is no separate reconstruction of a fine graph from the quotient before training.

The graph model replaces the ordinary deepest encoder–decoder pathway with a collector, sparse recurrent graph, and multiscale graph-to-decoder readout. The encoder produces feature maps

$$s_0, s_1, s_2, s_3, s_4, s_5$$

at spatial resolutions

$$256,\ 128,\ 64,\ 32,\ 16,\ 8$$

with channel dimensions

$$64,\ 128,\ 256,\ 512,\ 512,\ 512,$$

respectively. The graph-state width is $C = 32$.

The sparse forward operators are

$$1 \to 2, \qquad 1 \to 3, \qquad 2 \to 3,$$

with recurrent feedback

$$3 \to 2, \qquad 3 \to 1.$$

Sparse operators mix across graph nodes but not across feature channels. For an allowed directed sparse operator,

$$Y_{b,u,c} = b_u + \sum_{v:\, v \to u \text{ allowed}} \theta_{uv} X_{b,v,c},$$

where θ_{uv} is the learned scalar associated with an allowed graph connection and b_u is the learned node bias. Absent graph connections carry no sparse edge parameter.

Collector. Each encoder level is projected to graph width by a 1×1 convolution and flattened into node features. The allocation over the 4,802 nodes of the first color class is

s_3 :	32×32	1024
s_4 :	16×16	256
s_5 :	8×8	64
s_2 :	$64 \times 64 \to 32 \times 64$	2048
s_1 :	$128 \times 128 \to 30 \times 32$	960
s_0 :	$256 \times 256 \to 18 \times 25$	450

in the order

$$[s_3, s_4, s_5, s_2, s_1, s_0],$$

summing exactly to 4,802 graph nodes.

Recurrent graph update. Let a_1, a_2, a_3 denote the states of the three color classes. The collector initializes

$$a_1^0 = \mathrm{LN}_1(\mathrm{Collector}(s_0, \ldots, s_5)), \qquad a_3 \leftarrow 0.$$

Each of five recurrent passes then applies

$$a_2 = \mathrm{GELU}(\mathrm{LN}_2(W_{12} a_1 + \mathrm{LN}_{fb}(W_{32} a_3))),$$

$$a_1 = a_1^0 + \mathrm{LN}_{fb1}(W_{31} a_3),$$

and

$$a_3 = \mathrm{LN}_3(W_{23} a_2 + W_{13} a_1).$$

When enabled, supernode attention is applied immediately after the corresponding population update.

Attention is defined only for non-empty same-color groups under the fixed supernode assignment. The median non-empty group contains 3 vertices in each of the first two populations and 4 in the third, with means of 4.9, 4.8, and 7.1, respectively. Pooled over all 3,001 non-empty color–supernode groups, 84.3% contain eight vertices or fewer. The largest groups in the three populations contain 94, 102, and 137 vertices. Forty-four of the 3,045 possible color–supernode slots are empty and therefore carry no attention block.

The update order is recurrent rather than a static feed-forward ordering. In particular, a_2 is computed before the $3 \to 1$ feedback updates a_1, so the effect of that feedback reaches a_2 on the following recurrent pass. The implementation therefore selects one concrete causal schedule for the three chromatic phase classes. The coloring fixes the minimum number of phase classes required by the structural dependency graph; it does not specify a unique temporal ordering. Dependencies may continue across recurrent passes, so chromatic causal depth does not bound total computational depth.

Twin path and decoder coupling. The graph state is mapped back into decoder features at six scales. The two deepest outputs are learned projections over all $n_3 = 7{,}203$ nodes and are read from intermediate recurrent passes:

$$t_5 : \ 8 \times 8 = 64 \quad \text{from pass 3,}$$

$$t_4 : \ 16 \times 16 = 256 \quad \text{from pass 4.}$$

The remaining four outputs are taken from the final recurrent pass:

$$\begin{array}{lll} t_3 : & \to 32 \times 32 & 1024 \\ t_2 : & \to 32 \times 64 \to 64 \times 64 & 2048 \\ t_1 : & \to 32 \times 64 \to 128 \times 128 & 2048 \\ t_0 : & \to 32 \times 64 \to 256 \times 256 & 2048. \end{array}$$

The graph readout supplies the deep semantic pathway. At shallower levels the ordinary encoder skips are retained and modulated by the corresponding graph outputs.

Variant without attention. A variant removes only the three supernode-attention modules; everything else is unchanged. It was run once.

Component	Graph+attention	Graph without attention
Encoder	26,108,928	26,108,928
Decoder	46,829,699	46,829,699
Twin path	2,370,752	2,370,752
Sparse graph	579,248	579,248
Collector	63,680	63,680
Supernode attention	12,864	0
Total	75,965,171	75,952,307

Table S2: Parameter accounting for the two graph-model variants. The final substrate fixes the sparse routing pattern; training learns the parameters carried by the allowed sparse operators. The matched U-Net baseline contains 87,103,619 parameters.

Parameter counts.

Training and checkpoint selection. All graph runs are trained from scratch for a fixed 50 epochs with AdamW, batch size 2, gradient clipping at norm 1.0, and a cosine learning-rate schedule decaying to zero. There is no early stopping.

Parameters in the encoder, decoder, and twin path use the base learning rate 3×10^{-4} with weight decay 10^{-4}. The sparse-graph-plus- collector group uses a $10\times$ learning-rate multiplier,

$$3 \times 10^{-3},$$

with weight decay 10^{-5}. The U-Net comparison should therefore be interpreted as an architecture-level computational comparison rather than as a topology-only ablation.

Checkpoint selection follows a rule fixed before the final runs. Validation loss is rounded to four decimal places; after epoch 40, the first sequence of three consecutive epochs with the same rounded value is identified, and the earliest checkpoint in that sequence is reported. All runs nevertheless continue through the full 50 epochs.

For the three graph-plus-attention runs, this rule selects

$$\text{seed 0: epoch 45,} \qquad \text{seed 1: epoch 44,} \qquad \text{seed 2: epoch 45.}$$

For the three matched U-Net runs, the same rule selects

$$\text{seed 0: epoch 47,} \qquad \text{seed 1: epoch 46,} \qquad \text{seed 2: epoch 46.}$$

The single no-attention mechanism-check run selects epoch 47.

The pixel loss is

$$\mathcal{L}_{\text{pix}} = 5 \, \|(1 - m) \odot (\hat{x} - x)\|_1 + \|m \odot (\hat{x} - x)\|_1,$$

with the observed-region term applied to the raw network output. A VGG16 perceptual loss on the hidden region is added with weight 0.1.

Three-seed graph evaluation. The primary graph evaluation uses benchmark masks with target coverage sampled from $U(0.02, 0.70)$. Each metric is computed within six coverage buckets and the six bucket values are then averaged. Thus the reported value is a mean of bucket-level metrics rather than a direct per-image aggregate.

Run	L1 (%)↓	PSNR↑	SSIM↑	LPIPS↓
Seed 0 (ep45)	1.4506	29.826	0.90403	0.09329
Seed 1 (ep44)	1.4539	29.823	0.90388	0.09349
Seed 2 (ep45)	1.4554	29.810	0.90369	0.09391
Mean	1.4533	29.820	0.90387	0.09356
SD	0.0024	0.008	0.00017	0.00032

Table S3: Seed-level performance of the graph-plus-attention model on the benchmark mask distribution. Values are means over the six coverage buckets on the held-out reporting split. FID is omitted because it has not been recomputed for the new three-seed graph experiment.

The same three graph-plus-attention checkpoints were also evaluated on the training-mask distribution. The corresponding results are

Run	L1 (%)	PSNR	SSIM	LPIPS
Seed 0 (ep45)	1.4099	29.352	0.90553	0.09179
Seed 1 (ep44)	1.4152	29.346	0.90540	0.09200
Seed 2 (ep45)	1.4154	29.343	0.90524	0.09243
Mean	1.4135	29.347	0.90539	0.09207
SD	0.0031	0.005	0.00015	0.00033

The training-band aggregate should not be interpreted as an improvement over the benchmark aggregate, because the two mask distributions populate the coverage buckets differently. In the four central coverage buckets that are well populated under both distributions, L1 differs by less than 0.01 percentage points and PSNR by less than 0.03 dB across the measured runs. In this toy example, a single run with the attention blocks removed performed about the same as the attention-equipped model.

The matched U-Net remains the principal architectural baseline and is evaluated under the benchmark-mask protocol. No U-Net result on the training-mask distribution is inferred here. The inpainting experiment therefore supports the narrower conclusion that the final fine graph can be instantiated as a sparse recurrent computational substrate and trained reproducibly end to end; it does not isolate the contribution of graph topology alone.

S1.6 Network-topology audit definitions and null models

We evaluate the generated graphs using a topology audit designed to test whether a graph reproduces the main graph-theoretic signatures commonly reported for structural brain networks. The audit consists of nine scored criteria, together with unscored descriptive measures: box-covering scaling. Unless stated otherwise, each scored comparison is evaluated against degree-preserving null graphs generated by degree-preserving edge randomization, so that the result is not merely a consequence of the degree sequence [5, 6]. Enrichment criteria (clustering, modularity, local efficiency, and the significance clauses of the rich-club test) require the empirical value to exceed the null ensemble at $p < 0.05$, i.e. above the 95th percentile of the null distribution.

The descriptive box-covering and broad-degree checks are reported to characterize the graph family, but they are not counted among the nine scored quotient-level criteria. This is especially important at the dense 1,015-node quotient scale, where graph distances are too compressed for box-covering fits to be interpreted in the same way as on the large sparse parent graph.

1. Clustering enrichment. Brain networks are locally clustered: neighbors of a node tend to be connected to one another more often than in a comparable random graph. We therefore require clustering, or transitivity, to be significantly larger than the degree-preserving null ensemble. This captures the segregated local structure emphasized in small-world network theory [7]. Clustering enrichment is not by itself a small-world criterion.

Rather, it measures the local-segregation component that distinguishes small-world networks from ordinary random graphs. A graph is small-world only when this local clustering is combined with near-random path length.

2. Path length near random. Small-world organization also requires short global paths. We compare the average shortest-path length of the graph to that of matched random graphs and require it to remain close to random. Operationally, the audit uses a tolerance of approximately 25 percent above the random-graph path length. This prevents graphs from passing merely because they are highly clustered but globally lattice-like. Shortest-path quantities were computed exactly on the largest connected component using sparse all-pairs shortest paths. We used SciPy's `csgraph.shortest_path` routine with Dijkstra mode on the unweighted, undirected adjacency matrix. Average path length was computed as the mean of all finite nonzero shortest-path distances. Global efficiency was computed from the same distance matrix by averaging reciprocal distances, with unreachable pairs contributing zero.

3. Small-world index. The small-world index combines the preceding two effects: clustering should be much larger than random, while path length should remain near random. We use the Humphries–Gurney statistic

$$\sigma = \frac{T/T_{\text{null}}}{L/L_{\text{null}}},$$

where T is transitivity and L is average path length. The graph is small-world when $\sigma > 1$ [8]. We compute the statistic against two null conventions: Erdős–Rényi $G(n, m)$ graphs (σ_{ER}, the scored criterion) and degree-preserving edge-swap graphs (σ_{dp}). Both are reported; every graph audited in the main article satisfies $\sigma > 1$ under both.

4. Modularity enrichment. Modularity enrichment measures whether the graph separates into communities more strongly than expected from its degree sequence alone. After community detection, each node i is assigned a community label c_i. For an undirected unweighted graph with adjacency matrix A, degrees k_i, and total edge count m, the modularity of a partition is

$$Q = \frac{1}{2m} \sum_{i,j} \left(A_{ij} - \frac{k_i\, k_j}{2m} \right) \mathbf{1}\{c_i = c_j\}.$$

Here A_{ij} records whether nodes i and j are connected, while $k_i k_j/(2m)$ is the expected adjacency contribution between the same two nodes under a degree-preserving configuration-model null. The indicator $\mathbf{1}\{c_i = c_j\}$ restricts the sum to node pairs assigned to the same community. Thus Q is large when communities contain more internal connectivity than would be expected from node degrees alone [9]. In our audit, modularity enrichment passes only when the detected partition has nontrivial modularity, $Q > 0.3$, and the observed value is significant against degree-preserving null graphs.

The community labels c_i are obtained by first applying a community detection algorithm to the graph. In our audit, we use Louvain community detection, a modularity-optimization

heuristic in the same community-detection tradition as earlier large-network methods [10, 2]. Thus c_i denotes the Louvain community assigned to node i.

5. Global efficiency near random. Global efficiency measures how efficiently information can travel across the whole graph. If d_{ij} denotes the shortest-path distance between nodes i and j, then

$$E_{\text{glob}} = \frac{1}{n(n-1)} \sum_{i \neq j} \frac{1}{d_{ij}}.$$

Thus short paths contribute strongly, long paths contribute weakly, and disconnected pairs contribute zero. We compare global efficiency with matched degree-preserving null graphs. The criterion is not that global efficiency should be maximized, but that it should remain close to random: brain-like networks combine efficient long-range communication with non-random local and modular structure. In our audit, this criterion passes when $E_{\text{glob}}/E_{\text{glob,null}}$ lies in the range $[0.8, 1.0)$ (near but below one) [11, 12].

6. Local efficiency enrichment. Local efficiency measures how efficiently the neighbors of a node can communicate with one another after that node is removed. For each node i, let $N(i)$ be its neighborhood and let G_i be the subgraph induced by $N(i)$. The local efficiency of the whole graph is

$$E_{\text{loc}}(G) = \frac{1}{n} \sum_{i=1}^{n} E_{\text{glob}}(G_i),$$

where

$$E_{\text{glob}}(G_i) = \frac{1}{k_i(k_i-1)} \sum_{\substack{j,h \in N(i) \\ j \neq h}} \frac{1}{d_{jh}^{(i)}}.$$

Here k_i is the degree of node i, and $d_{jh}^{(i)}$ is the shortest-path distance between neighbors j and h inside the neighbor-induced subgraph G_i. If $k_i < 2$, the contribution of node i is taken to be zero. Thus local efficiency is high when the neighbors of a node remain well connected even without that node. We require local efficiency to be enriched relative to degree-preserving null networks, indicating more local fault-tolerant neighborhood structure than expected from the degree sequence alone [11].

7. Connector hubs. Connectomes contain hubs that link multiple communities. After community detection, let c_i be the community of node i, let k_i be its total degree, and let k_{is} be the number of edges from node i to community s. The participation coefficient of node i is

$$P_i = 1 - \sum_{s} \left(\frac{k_{is}}{k_i} \right)^2.$$

This quantity is small when most edges of node i remain inside one community, and large when its edges are distributed across several communities. We also compute the within-module degree score

$$z_i = \frac{k_i^{\text{in}} - \mu_{c_i}}{\sigma_{c_i}},$$

where k_i^{in} is the number of edges from node i to nodes inside its own community, and μ_{c_i} and σ_{c_i} are the mean and standard deviation of within-community degree among nodes in the same community as i. Following the Guimera–Amaral cartographic classification, we call node i a connector hub when

$$z_i \geq 2.5 \qquad \text{and} \qquad 0.30 < P_i \leq 0.75.$$

Thus connector hubs are nodes that are hub-like inside their own module while also linking substantially to other modules. The presence of at least one such connector hub is used as evidence of cross-module integration [13].

8. Rich-club organization. Rich-club organization tests whether high-degree nodes are more densely interconnected with one another than expected under a null model. For a degree threshold k, let $N_{>k}$ be the number of nodes with degree greater than k, and let $E_{>k}$ be the number of edges among those nodes. The rich-club coefficient is

$$R(k) = \frac{2E_{>k}}{N_{>k}(N_{>k} - 1)}.$$

This is the edge density of the subgraph induced by nodes whose degree exceeds k. To test whether this density is larger than expected from the degree sequence alone, we normalize by a degree-preserving null ensemble:

$$\rho(k) = \frac{R(k)}{\langle R_{\text{null}}(k) \rangle}.$$

A value $\rho(k) > 1$ means that high-degree nodes are more interconnected than in matched null graphs. In our audit, rich-club organization passes when $R_{\text{norm}}(k) > 1$ with statistical significance over a run of at least three consecutive degree thresholds, indicating a supported high-degree backbone [14, 15].

9. Multi-resolution VI stability. This criterion follows the multi-scale brain-network view that community structure should be examined across topological scales rather than at a single fixed resolution [16]; variation of information provides the partition-distance measure used to identify reproducible scales [17].

Hierarchical modular organization is tested by applying Louvain community detection at several resolution parameters γ. The resolution parameter controls the scale of the detected communities: smaller values favor larger communities, while larger values favor smaller communities. At each fixed value of γ, we repeat Louvain several times with different random seeds. These repeats test whether the community structure at that particular resolution is reproducible.

We compare the repeated community partitions using variation of information,

$$\text{VI}(\mathcal{P}, \mathcal{Q}) = H(\mathcal{P}) + H(\mathcal{Q}) - 2I(\mathcal{P}; \mathcal{Q}),$$

where $\mathcal{P}$ and $\mathcal{Q}$ are two detected community partitions, H denotes partition entropy, and I denotes mutual information. Low VI means that repeated runs found similar community

assignments. We then look across the tested values of γ for local minima of the mean VI. A local minimum identifies a resolution at which the community structure is especially reproducible. We count such a scale only when the corresponding modularity is also nontrivial, with $Q > 0.3$ [17, 18]. The criterion passes when at least one VI-stable modular scale is present.

10. Box-covering scaling (descriptive). Box covering asks how many boxes of a given size are needed to cover a graph. For a box size l_B, let $N_B(l_B)$ be the number of boxes in a covering; where the count follows a power law over a range of sizes, the box-covering dimension is the negative slope

$$d_B = -\frac{d \log N_B(l_B)}{d \log l_B},$$

fitted by least squares on $\log N_B$ against $\log l_B$ [3, 4]. Exact minimum box covering is NP-hard, so the count is always obtained by an approximation. This work uses two different approximations in two different roles, and they are not interchangeable; both are given below.

(a) In the audit, as a measurement. Box counts are obtained by the maximum-excluded-mass burning (MEMB) algorithm of Song et al. [4], run on the exact all-pairs shortest-path matrix d of the largest connected component. For a burning radius r_B:

1. Initialise every vertex as uncovered and as a non-centre. For each vertex u, set its excluded mass $\text{mass}(u) = |\{v : d(u, v) \leq r_B\}|$.
2. While some vertex is neither covered nor a centre:
   (a) Among all non-centre vertices — including those already covered, following Song's step (ii) — select the vertex c of maximum excluded mass.
   (b) If that maximum is zero, give each remaining uncovered non-centre vertex its own box and stop.
   (c) Mark c a centre. Assign to its box every uncovered vertex v with $d(c, v) \leq r_B$, mark those vertices covered, and increment the box count.
   (d) For every vertex u, reduce $\text{mass}(u)$ by the number of newly covered vertices within r_B of u; set the mass of every centre to zero.
3. Report $N_B(l_B)$ as the resulting box count, with box size $l_B = 2r_B$.

The procedure is deterministic: no random choices enter it. Burning within radius r_B bounds pairwise distances inside a box by $2r_B$, which is why l_B is defined as $2r_B$; Song's text writes $l_B = 2r_B + 1$ under a strict distance bound, and we take the offset that reproduces the published benchmark values more closely. Step 3 is repeated for $r_B = 1, \ldots, \lfloor \text{diam}/2 \rfloor$, so the audited scales are even. Scales at which the graph is covered by a single box are discarded, the remaining points are fitted in log–log to give d_B, and an exponential $N_B \propto e^{-\beta l_B}$ is fitted to the same points for comparison. This is the only place d_B is measured. It is descriptive and is not one of the nine scored criteria: it is reported to characterize the graph rather than to assert fractality, and only where the graph affords enough distance range for a fit.

(b) In the construction, as a mechanism. The fractal-targeted sparsification stage covers the graph in boxes in order to impose a property rather than to measure one: edges of low triangle support are removed until the stage's own estimate of d_B falls to its threshold. Its covering is a randomized greedy approximation, not MEMB. For a scale l_B:

1. Repeat for 30 independent trials:
   (a) Draw a uniformly random permutation of the vertices; mark all vertices uncovered and set the box count to zero.
   (b) Scan the permutation in order. Skip a vertex if it is already covered. Otherwise take it as a centre c, mark covered every vertex v with $d(c, v) \leq l_B$, and increment the box count.
   (c) Stop the scan once every vertex is covered.
2. Take $N_B(l_B)$ to be the smallest box count over the 30 trials.

This is repeated over the scales $\{2, 3, 4, 5, 6, 8, 10, 12\}$, restricted to those not exceeding the graph diameter; single-box scales are discarded and the remaining points fitted in log–log as above. Two differences from (a) are worth making explicit. Centres are taken in random order rather than by excluded mass, so the covering is a best-of-30 heuristic rather than a deterministic construction. And the ball is taken at radius l_B itself, not at $l_B/2$: for the same numeric label, a box here has twice the radius of a box in (a). The bridge stage uses a covering of the same greedy kind over the scales $\{3, 4, 5, 6, 7\}$, differing in taking the strict ball $d(c, v) < l_B$ and in using 50 trials rather than 30; its box labels are used solely as a multiscale locality filter on candidate edges — no dimension is fitted there.

Because the centre rule, the meaning of l_B, and the scale set all differ, the threshold at which sparsification halts and the dimension the audit subsequently reports are not the same quantity, and the two should not be read as a target and a miss.

S2 Supplementary Results

S2.1 Parent-graph audits

The local construction, the original integrated parent, and the final relocation substrate are distinct fine-scale graphs. The inverse procedure changes edge placement under the fixed 1,015-supernode causal partition; the final relocation substrate is the graph used directly by the recurrent neural model.

S2.1.1 Local parent graph

The local construction gives

$$n = 16{,}807, \qquad m = 277{,}319.$$

Against tripartite degree-preserving nulls it passes seven of the nine scored criteria. The two failures are the explicitly integrative criteria: near-random path length and near-random global efficiency. Clustering, small-world sigma, modularity, local efficiency, connector hubs, rich-club organization, and multi-resolution modular stability all pass.

S2.1.2 Original integrated parent graph

Exploratory; not part of the main article's reported results. The original integrated parent has

$$n = 16{,}807, \qquad m = 337{,}983, \qquad \rho = 0.002393,$$

with color-class sizes 4,802/4,802/7,203. Against 1,000 tripartite degree-preserving nulls it passes all nine scored criteria. Its principal values are

$$C = 0.0366, \quad T = 0.0317, \quad Q = 0.4605, \quad E_{\mathrm{loc}} = 0.1208,$$

$$L = 3.7871, \quad E_{\mathrm{glob}} = 0.2844, \quad \sigma_{\mathrm{dp}} = 1.68, \quad \sigma_{\mathrm{ER}} = 10.24,$$

with assortativity +0.2542. The longest significant rich-club run is 290. The descriptive box-covering fit is

$$d_B = 4.03, \qquad R^2 = 0.94,$$

and is not part of the nine scored criteria.

S2.1.3 Final relocation substrate

Exploratory; not part of the main article's reported results.

The final relocation substrate has

$$n = 16{,}807, \qquad m = 337{,}980.$$

It shares 50.3% of its edges with the original integrated parent, for edge-set Jaccard similarity 0.336. Its local and mesoscale organization is substantially stronger:

$$C = 0.2281, \quad T = 0.3089, \quad E_{\mathrm{loc}} = 0.3497, \quad Q = 0.8104,$$

with assortativity +0.5802,

$$\sigma_{\mathrm{dp}} = 15.37, \qquad \sigma_{\mathrm{ER}} = 83.80.$$

Its average path length is 4.5114 and global efficiency is 0.2439. Against 100 tripartite degree-preserving nulls it passes seven of the nine criteria, failing only the tests requiring path length and global efficiency to remain near their randomized counterparts. This is a near-randomness failure rather than an absolute long-path regime: approximately 93% of pairs are within six hops, the maximum observed hop distance is 17, and a same-size ring lattice has average path length about 210.6. The descriptive box-covering fit is

$$d_B = 3.625, \qquad R^2 = 0.952.$$

The recurrent neural model uses this final relocation substrate directly.

Table S4: Nine-criterion topology audit for the local quotient (n = 1,015, $|E|$ = 67,094) against the Budapest Reference Connectome. Parenthetical values are ratios to degree-preserving null means. Both graphs pass all nine scored criteria.

Criterion	Local quotient	Budapest	Verdict
Clustering enriched	$C = 0.515\,(2.46\times)$	3.04×	PASS
Path length near random	APL $= 1.97\,(1.05\times)$	1.19×	PASS
Small-worldness	$\sigma_{\mathrm{ER}} = 3.49$ (scored); $\sigma_{\mathrm{dp}} = 2.30$	$\sigma_{\mathrm{ER}} = 3.41$ (scored); $\sigma_{\mathrm{dp}} = 2.22$	PASS
Modularity enriched	$Q = 0.524\,(8.24\times)$	9.1×	PASS
Global efficiency near random	$E_{\mathrm{glob}} = 0.548\,(0.97\times)$	0.90×	PASS
Local efficiency enriched	$E_{\mathrm{loc}} = 0.737\,(1.24\times)$	1.38×	PASS
Connector hubs	4 hubs	4 hubs	PASS
Rich-club organization	294 sig. thresholds; run 177	212 sig. thresholds; run 212	PASS
Multi-resolution VI stability	stable partition	stable partition	PASS
Score: 9/9 **for both graphs.**			

S2.2 Extended local-quotient analyses

S2.2.1 Local quotient topology audit

After coarsening, the local quotient passes the same nine scored topology criteria used below for the final quotient: clustering enrichment, near-random path length, degree-preserving small-worldness, modularity enrichment, near-random global efficiency, local efficiency enrichment, connector hubs, rich-club organization, and multi-resolution VI-stable modularity.

The local-quotient audit uses 100 degree-preserving nulls for the main metric criteria and 1,000 nulls for rich-club testing. Rich-club results are reported as the number of significant degree thresholds together with the longest consecutive significant run, both derived from the same 1,000-null curves. Threshold counts are restricted to degree thresholds k for which at least 20 nodes have degree $> k$, with significance assessed at $p < 0.05$. For connector hubs, the reported value is the participation-based hub count, and × denotes a multiple of the corresponding random-null value.

Budapest lies deeper within the enrichment criteria for clustering (3.04× versus 2.46×), modularity (9.1× versus 8.24×), and local efficiency (1.38× versus 1.24×). The local quotient has a similar degree-preserving small-worldness value ($\sigma_{\mathrm{dp}} = 2.30$ versus 2.22) and also exhibits statistically significant rich-club organization, although the detailed threshold profile differs between the two graphs. Thus the 9/9 score should be read as agreement at the level of the stated topological criteria rather than equality of the underlying numerical statistics.

We therefore treat the quotient audit as a Budapest-scale coarse-graph result, not as proof that all nine properties are intrinsic to the fine computational substrate. The parent audit in Supplementary Section S2.1 is the stronger substrate-level test. The difference between the parent score of 7/9 and the quotient score of 9/9 shows that coarsening supplies the missing

Measure	Local	Budapest
$\lambda_{\max}(A)/\langle d\rangle$	1.380	1.285
$\log \tau(G)/n$	4.739	4.774
Fiedler value λ_2	2.964	2.654
Hoffman bound	3.96	7.91
Spectral (Fiedler) bisection cut/m	0.4472	0.0123
Maximum matching, normalized	0.999	0.999
Fractional matching	1.000	1.000
Fractional vertex cover	0.500	0.500
Vertex cover, 2-approx. normalized	0.974	0.982
Minimum spanning forest	1014	1014

Table S5: Algebraic and combinatorial comparison between the local quotient and the Budapest Reference Connectome. Several quantities are close, including spanning-tree entropy and the Fiedler value. Matching, fractional matching, fractional vertex cover, and minimum spanning forest size are nearly saturated at this density and are reported for completeness rather than interpreted as strong independent evidence.

integration in the local construction.

S2.2.2 Independent local-quotient graph battery

Tables S5–S8 summarize the independent graph battery for the local quotient. These quantities were not used to construct it. They include algebraic and combinatorial measures, graph-complexity measures, spectral quantities, network-controllability statistics, and clique-census measures.

The local quotient is close to Budapest on spanning-tree entropy and Fiedler value. The strongest mismatch is the spectral bisection cut; as shown in the main article, the local quotient does separate cleanly along its own communities (a community-aligned cut of 6.27%), but its Fiedler direction does not follow them. The low Hoffman bound is not simply a failure: it is the spectral signature of the local graph remaining closer to the underlying chromatic construction.

The local variant remains informative because several bulk spectral and resistance-based quantities, especially graph energy and Kirchhoff index, already lie close to Budapest. The simple construction therefore captures substantial parts of the large-scale graph regime without the later fixed-partition inverse relocation.

Measure	Local	Budapest
Off-diagonal complexity	5.446	5.406
Degree entropy	5.283	5.405
Degree-pair entropy	9.733	9.899
Kirchhoff index	11,228	10,849
$\log_{10}$ Estrada index	79.25	77.72
Graph energy	5,887	5,778
Graph energy per node	5.80	5.69

Table S6: Graph-complexity and spectral comparison. The local quotient is close to Budapest on entropy-like measures, the Kirchhoff index, and graph energy. Because the local quotient is slightly sparser than Budapest (67,094 versus 70,654 edges), the Kirchhoff-index and graph-energy comparisons should be read as qualitative rather than density-matched claims.

Measure	Local	Budapest
Average controllability, mean	1.094	1.100
Average controllability, CV	0.137	0.101
Modal controllability, mean	0.9961	0.9957
$\mathrm{corr}(d, \mathrm{avgctrl})$	$+0.818$	$+0.875$
$\mathrm{corr}(d, \mathrm{modalctrl})$	-1.000	-1.000

Table S7: Network-controllability comparison. The local quotient reproduces the qualitative controllability profile of Budapest: high-degree nodes have high average controllability, while low-degree or peripheral nodes have high modal controllability.

Measure	Local	Budapest
Clique number ω	43	64
Total maximal cliques	5.03 M	15.4 M
Maximal-clique histogram peak	$k = 30$	$k = 40$
Participation Gini	0.815	0.812
Participation-degree correlation	0.742	0.779
High-participation nodes	62 (6.1%)	64 (6.3%)

Table S8: Higher-order clique comparison between the local quotient and the Budapest Reference Connectome. The local quotient reaches $\omega = 43$ versus Budapest's 64, and its maximal-clique-size distribution peaks at $k = 30$, close to Budapest's $k = 40$. The participation Gini, the participation-degree correlation, and the fraction of high-participation nodes are all close to Budapest.

The clique census identifies a remaining higher-order difference. The local quotient reaches $\omega = 43$, below Budapest's 64, even though its maximal-clique-size distribution peaks at $k = 30$, reasonably close to Budapest's $k = 40$. This reinforces the interpretation of the

local quotient as occupying the same broad mesoscale regime without reproducing the full higher-order closure of the reference connectome.

S2.3 Strict-coarsening obstruction and inverse relocation

Exploratory; not part of the main article's reported results.

This section documents the optional complex refinement summarized near the end of the main article. Its purpose is specifically to show what additional mechanism is required when fiber-bundle multiplicity and the strict quotient are targeted simultaneously; it is not part of the basic 8,000-node construction.

Under strict collapse, every fine edge whose endpoints lie in different supernodes contributes to the corresponding quotient relation; only intra-supernode edges disappear. For the fixed map π,

$$W_{ab} = \#\{\{u, v\} \in E : \pi(u) = a,\ \pi(v) = b\}, \qquad a \neq b.$$

The unchanged integrated parent yields 143,679 cross-supernode relations. The selected mesoscale target-support scaffold contains only 70,538 relations. Physically deleting the support of all omitted relations would remove 101,816 fine edges supporting 73,141 coarse relations, leaving 236,167 parent edges, three connected components, and minimum degree one. Thus the target support cannot be treated as a strict quotient of the unchanged parent.

The obstruction is not specific to this partition. Twenty-five alternative strict coarsenings were forced to exactly 1,015 groups. Representative results are shown in Table S9.

Map	Quotient edges	Q	C	APL
Budapest	70,654	0.5574	0.6696	2.219
Leiden RB, res. 40	60,513	0.288	0.275	1.882
Leiden CPM, 0.05	13,080	0.217	0.541	2.014
Leiden RB, res. 30	90,277	0.204	0.330	1.825
METIS balanced	84,729	0.196	0.389	1.863
Leiden RB, res. 20	141,633	0.124	0.373	1.725
Random balanced	234,240	0.040	0.516	1.545

Table S9: Representative members of the 25-map strict-coarsening sweep. No tested strict coarsening of the unchanged integrated parent reaches the Budapest modular regime.

The reported inverse construction therefore preserves the original community-derived causal partition and adjusts the fine substrate instead:

$$(G_{\text{int}}, \pi) \longrightarrow (G_{\text{final}}, \pi) \xrightarrow{\text{strict collapse}} H_{\text{final}}.$$

Every fine vertex remains in the same supernode. What changes is the pattern of fine causal-dependency edges within and between those fixed populations.

S2.3.1 Strict verification and bundle accounting

In the final relocation substrate, the 337,980 fine edges divide into

147,217 cross-supernode edges

and

190,763 intra-supernode edges,

which sum exactly to 337,980. Recomputing the quotient directly from the completed parent reproduces the saved final quotient edge set and weights exactly. No cross-supernode fine edge is omitted.

Bundle statistic	Final quotient	Budapest
Mean	2.071	2.033
Median	2.000	1.667
Maximum	155	154.9
Gini coefficient	0.306	0.305
Relations with weight > 18	41	40

Table S10: Fine-edge multiplicities underlying the final quotient. The match is distributional; no constructed quotient edge is assigned anatomical identity with a particular Budapest tract.

S2.4 Final strict quotient

Exploratory; not part of the main article's reported results.

The single final quotient reported in the paper has

$$n = 1{,}015, \qquad m = 71{,}098, \qquad \rho = 0.1382,$$

compared with Budapest's 1,015 nodes, 70,654 edges, and density 0.1373. The degree distributions are closely aligned ($KS = 0.0187$).

Quantity	Final quotient	Budapest
Modularity Q	0.4908	0.5574
Clustering C	0.4724	0.6696
Transitivity	0.4234	0.5578
Average path length	1.9811	2.2190
Diameter	4	5
Effective diameter	2.162	2.731
Mean degree	140.1	139.2
Median degree	130	127
Minimum degree	13	7
Maximum degree	420	466
Assortativity	+0.0826	+0.0215

Table S11: Principal scalar comparison between the final strict quotient and the Budapest Reference Connectome.

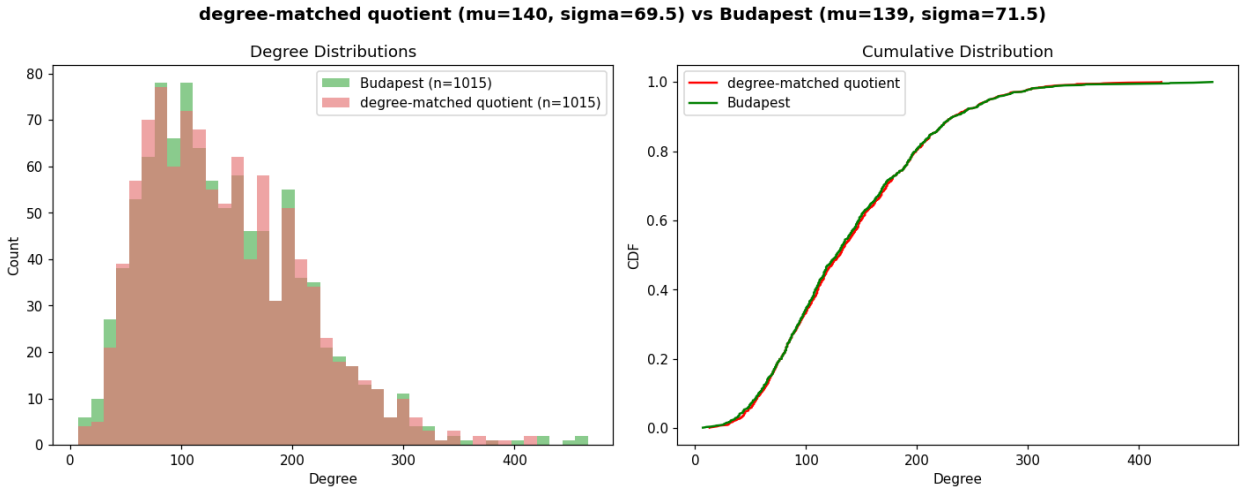


Figure S1: Degree comparison for the final strict quotient and Budapest. The final degree-profile adjustment produces a close distributional match ($KS = 0.0187$) without assigning anatomical correspondence between individual nodes.

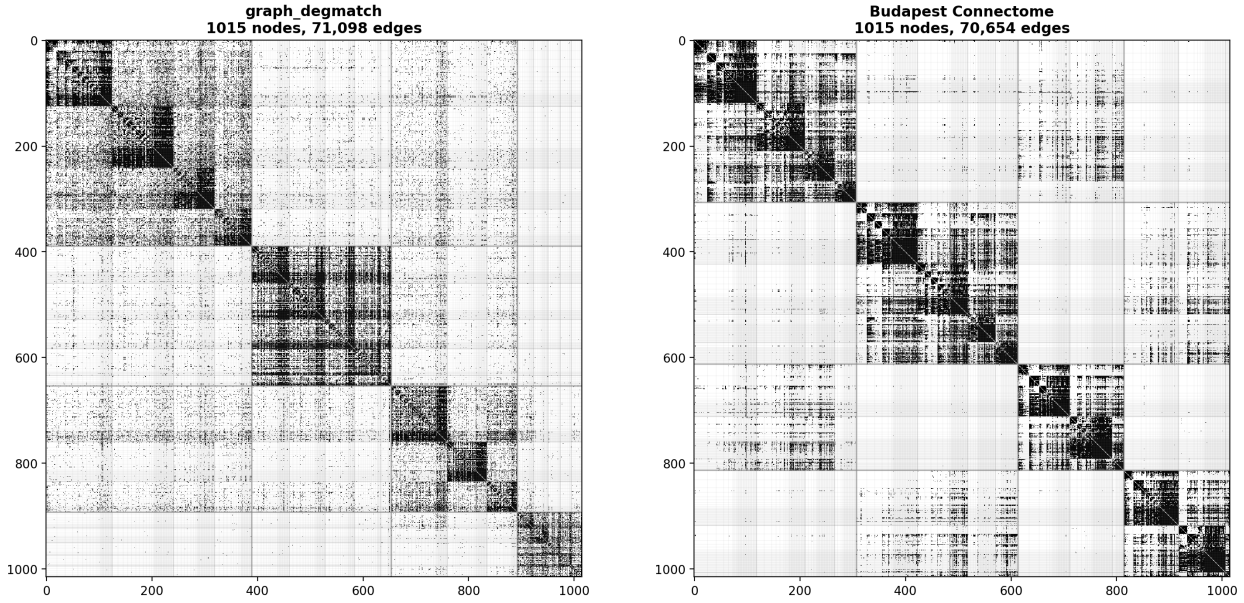


Figure S2: Community-ordered adjacency matrices of the final strict quotient (left) and the Budapest Reference Connectome (right), both with $n = 1{,}015$. The constructed graph is the complete strict collapse of the final relocation substrate under the fixed community-derived causal partition. Budapest shows a stronger bilateral large-scale organization that is absent from the construction.

S2.4.1 Nine-criterion audit

Against 1,000 degree-preserving nulls, the final quotient passes all nine scored criteria. The score denotes criterion-level agreement rather than numerical identity.

Criterion	Final quotient	Verdict
Clustering enriched	$C = 0.4724$, 2.198× null	PASS
Path length near random	$L = 1.9811$, 1.063× null	PASS
Small-worldness	$\sigma_{\mathrm{ER}} = 2.880$ (scored); $\sigma_{\mathrm{dp}} = 1.918$	PASS
Modularity enriched	$Q = 0.4908$, 8.00× null	PASS
Global efficiency near random	$E_{\mathrm{glob}} = 0.5492$, 0.966× null	PASS
Local efficiency enriched	$E_{\mathrm{loc}} = 0.7304$, 1.216× null	PASS
Connector hubs	participation-based connector structure present	PASS
Rich-club organization	272 significant thresholds; longest run 272	PASS
Multi-resolution VI stability	stable modular scale present	PASS
Overall score: 9/9**.**		9/9

Table S12: Nine-criterion audit of the final strict quotient. All comparisons use the stated degree-preserving null framework; the 9/9 result does not imply numerical identity with Budapest.

Descriptive box-covering scaling is not scored at this dense scale. Under the same descriptive check, neither the final quotient nor Budapest supports a clean fractal-scaling interpretation.

S2.4.2 Six static graph tests

The final quotient is also compared with Budapest on degree distribution, hop plot, singular-value scree, network value, triangle participation, and effective diameter. Its six-test mean normalized error is 0.3033, with degree-distribution $KS = 0.0187$. The degree and cumulative network-value profiles are close; the largest residual is spectral.

S2.4.3 Rich-club organization

Using 1,000 degree-preserving nulls, the ≥ 20-node threshold floor, and $p < 0.05$, the final quotient has 272 significant degree thresholds in one consecutive run of length 272. Budapest has 212 significant thresholds, also in one run.

S2.4.4 Complexity, spectral, and controllability diagnostics

These quantities were not individual fitting targets and are reported as descriptive comparisons.

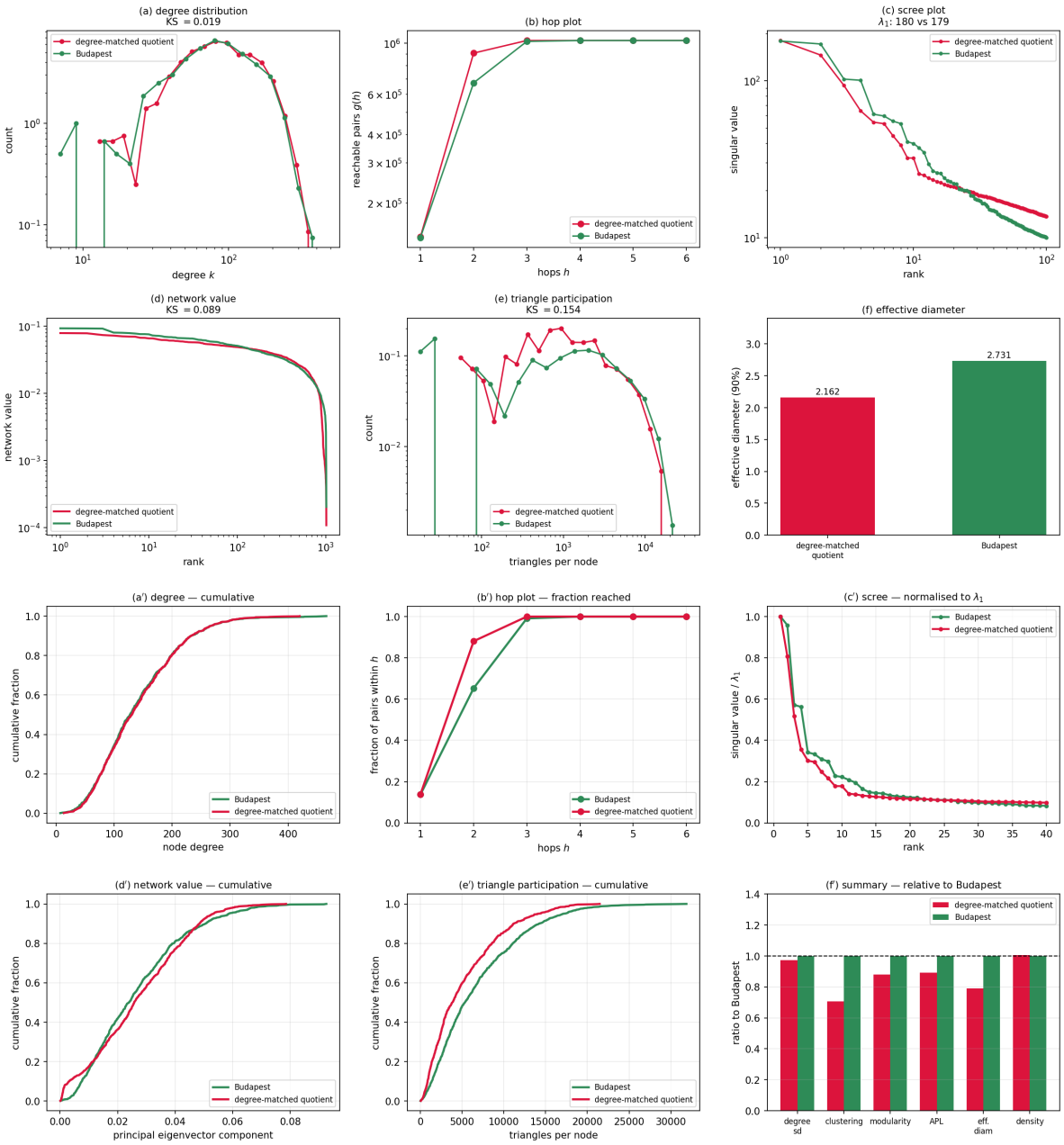


Figure S3: Six static graph tests comparing the final strict quotient (n = 1,015, m = 71,098) with Budapest (n = 1,015, m = 70,654). Rows 1–2 use the log–log convention and rows 3–4 show cumulative versions. The degree match is especially close; the principal residual difference is spectral.

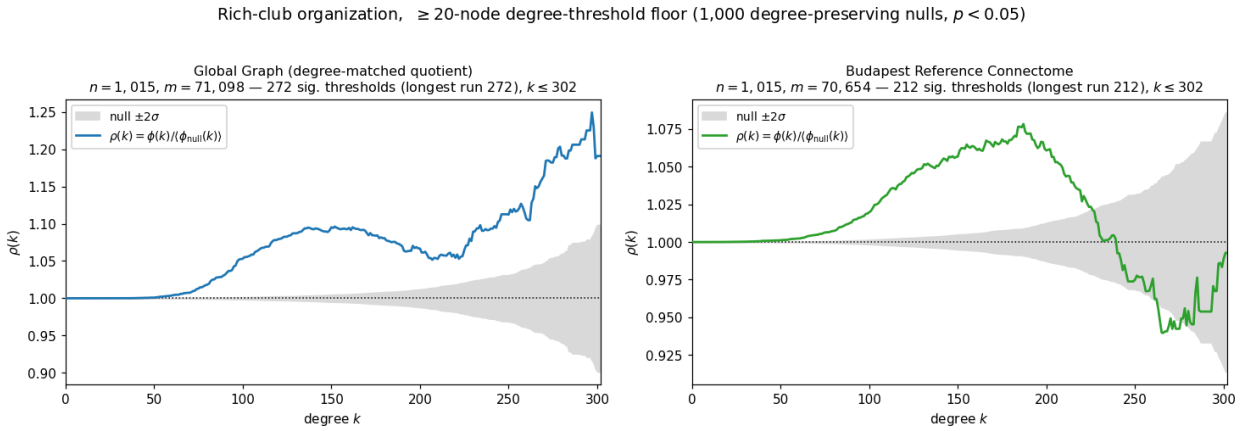


Figure S4: Normalized rich-club coefficient for the final strict quotient (left) and Budapest (right), each evaluated against 1,000 degree-preserving null graphs and restricted to thresholds retaining at least 20 nodes. The final quotient has a significant run of length 272, compared with 212 for Budapest.

Measure	Final quotient	Budapest
$\log_{10}$ Estrada index	78.34	77.72
Graph energy / n	7.009	5.693
Kirchhoff index / n^2	0.00985	0.01053
Off-diagonal complexity	5.318	5.406
Degree entropy	5.410	5.405
$\lambda_{\max}/\langle d\rangle$	1.288	1.285
Hoffman bound	6.612	7.908
Random-walk eigengap	0.0938	0.0197
$\log \tau(G)/n$	4.791	4.774
Average controllability	1.0932	1.1004

Table S13: Selected complexity, spectral, and controllability diagnostics. Resistance-based, spanning-tree, leading-eigenvalue, average-controllability, off-diagonal-complexity, and degree-entropy quantities are close, while the random-walk eigengap remains different.

The clearest unresolved large-scale feature is bilateral organization. The Budapest graph has a near-degenerate leading spectral pair $\lambda_1 = 178.9$, $\lambda_2 = 171.4$ and a top-level community pattern 307/306/201/201 that naturally separates into approximately balanced halves. The final quotient instead has top-level sizes 389/265/239/122 and no paired organization. The 1,015-node match is therefore distributional rather than anatomical; bilateral organization would require additional anatomical or spatial information not present in the substrate.

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